Chapter 5
Stock Valuation

Principles of Finance

LO1: Explain how stock prices depend on future dividends and dividend growth – Chapter 5

In this chapter, we delve into the core concepts of stock valuation. We'll explore how the prices of stocks are intrinsically linked to the anticipation of future dividends and the rate at which those dividends grow.

5.1
Common Stock Valuation

Uncertain Cash Flows

Valuing common stock is more complex than valuing a bond because the promised cash flows are uncertain.

Perpetual Life

The life of a stock is essentially perpetual, unlike a bond with a finite maturity.

Market's Required Rate of Return

It's difficult to determine the market's required rate of return, as it reflects a multitude of factors.

Cash Flows

1

Today

You are a student considering investing in a company's stock.

2

After a Year

The stock is expected to be worth SAR 70 and pay a dividend of SAR 10.5 per share.

3

Desired Return

You want to earn a 15% return on your investment.

How to calculate stock value?

> To determine how much you would pay for this stock today, you need to calculate the present value of the expected future benefits: the SAR 10.5 dividend and the SAR 70 you anticipate receiving from selling the stock after a year.

If the company promises to pay this SAR 10 dividend *forever*, this is the idea of a **perpetuity**. It's like a never-ending stream of payments.

Zero Growth Stock

If the company promises to pay this SAR 10 dividend *forever*? This is the idea of a **perpetuity**. It's like a never-ending stream of payments.

Example: Tasnee stock

Tasnee pays 3 SAR annual dividend.

Required Return

Required return: 12%

Formula

$$(P = \frac{D}{R})$$

Calculation

$$(P = \frac{3}{0.12} = 25, SAR)$$

Constant Growth Stock

The dividend growth model determines the current price of a stock as its dividend next period divided by the discount rate less the dividend growth rate.

Example: Mobily stock

Mobily pays 4 SAR annual dividend.

Growth and Return

Expected growth rate: 5% Required return: 15%

Formula

$$P_0 = \frac{D_0(1+g)}{R-g}$$

Calculation

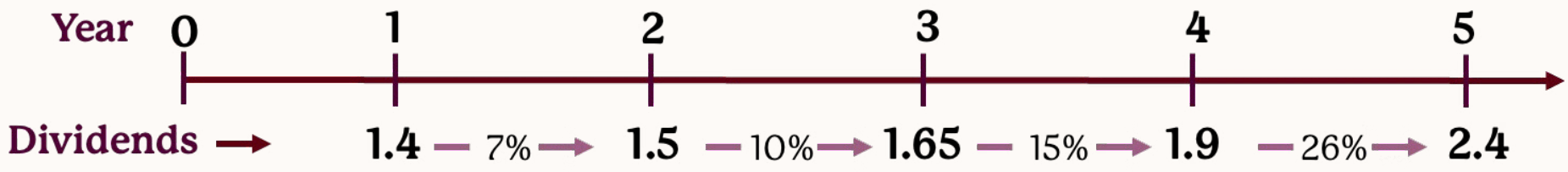
$$P_0 = \frac{4 \times (1+0.05)}{0.15-0.05} = 42, SAR$$

Nonconstant Growth: When Dividends Don't Follow the Rules

Sometimes, a company's dividend growth isn't consistent. It might grow quickly at first, then slow down to a more steady pace.

The nonconstant growth model helps us value these stocks. It considers both the initial rapid growth and the eventual stable growth, giving us a more accurate valuation.

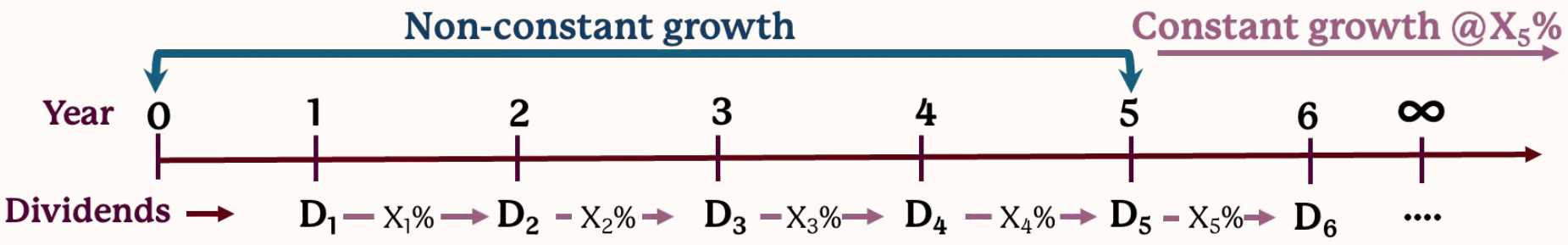
This model is especially useful for companies in their early stages or those going through major changes.



Two-stage Growth Model

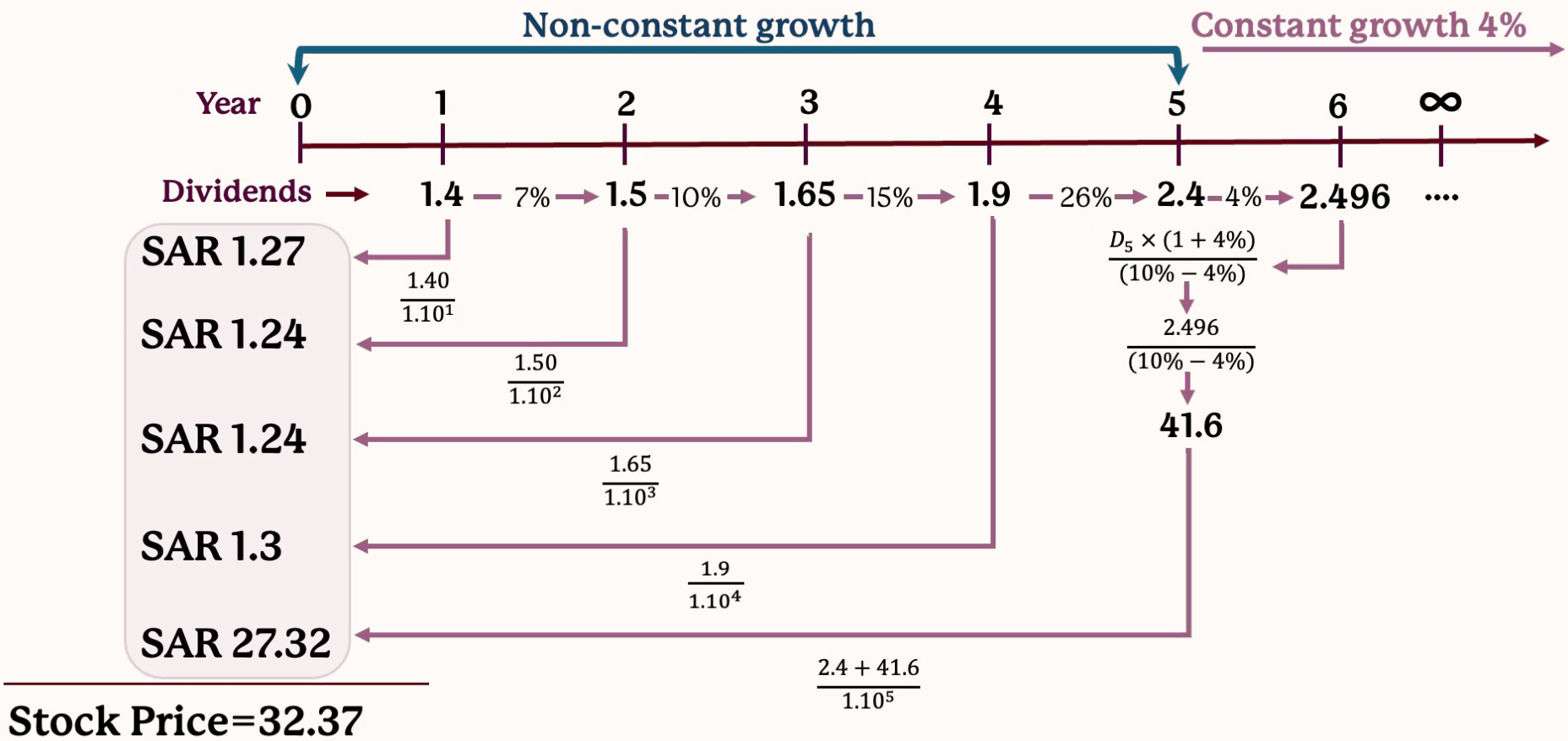
The two-stage growth model is a powerful tool for valuing companies with nonconstant dividend growth. It combines the constant growth model with a period of initial rapid growth, leading to a more accurate valuation.

This model is especially useful for companies in their early stages or going through significant changes where dividend growth is not yet predictable.



Two-Stage Growth Model: Step-by-Step

A company pays SAR 1.40, 1.50, 1.65, 1.90, and 2.40 dividends over five years. After that, dividends grow at a steady 4% forever. The discount rate is 10%.



Wrap Up: Understanding Stock Values

The Constant Growth Model

This model values stocks with consistent dividend growth.

The Nonconstant Growth Model

This model values stocks with periods of rapid and stable growth.

Understanding Stock Values

Stock value reflects investor expectations of future dividends and growth.

Key Takeaways

1. **Stock Price Drivers**
   Stock prices are linked to future dividends and their growth.

2. **Growth and Value**
   Higher growth rates can lead to higher stock prices.

3. **Dividend Appeal**
   Stable and growing dividends enhance stock attractiveness.

5.2
Stock Valuation

Principles of Finance

Show how to value stocks using multiples – Chapter 5

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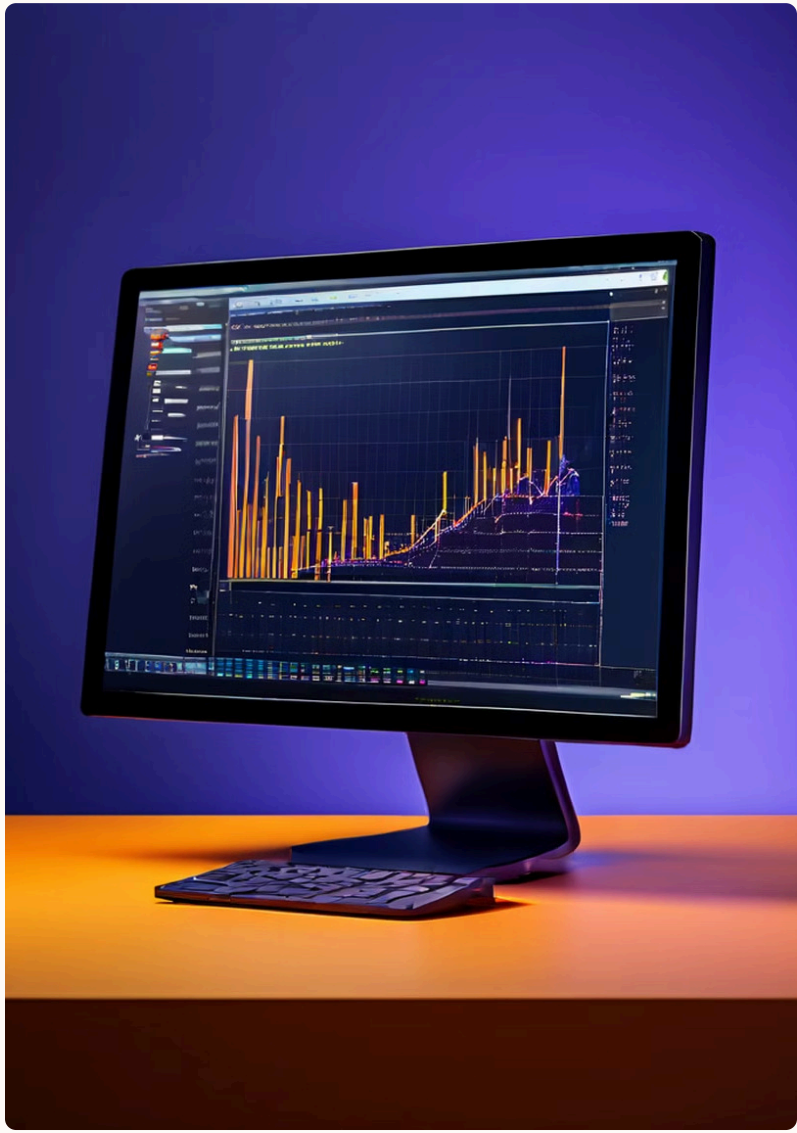


Stock Valuation Using Multiples

One challenge with valuing stocks based on dividends is that many companies don't pay them. If a company is profitable, you can use the price-to-earnings (P/E) ratio. This ratio compares the stock's price per share to its earnings per share (EPS) over the past year.

Price-Earnings (P/E) Ratio

The idea is to have some sort of benchmark PE ratio, which we then multiply by earnings to come up with a price. The benchmark PE ratio could come from a variety of sources (for example, based on similar companies, based on a company's own historical values). A PE ratio based on estimated future earnings is a forward PE ratio.

Jarir Bookstore's EPS

4 SAR

Industry Benchmark P/E Ratio

15

Formula

$$(\text{Stock Price} = \text{Benchmark P/E Ratio} \times \text{EPS})$$

Calculation

$$(15 \times 4 = 60, SAR)$$

Price-Sales (P/S) Ratio

The Price-Sales (P/S) Ratio is a valuation multiple that compares a company's market capitalization (its total equity value) to its annual revenue. It gives you a sense of how much investors are willing to pay for every SAR 1 of sales generated by the company.

1 Ninja Delivery App's Sales per Share

10 SAR

2 Industry Benchmark P/S Ratio

Jahiz P/S = 2

3 Formula

$$(\text{Stock Price} = \text{Benchmark P/S Ratio} \times \text{Sales per Share})$$

4 Calculation

$$(2 \times 10 = 20, SAR)$$

- **Higher P/S Ratio:** A higher P/S ratio suggests that investors are paying more for each SAR 1 of sales, indicating higher expectations for future growth, strong brand value, or a competitive market position.
- **Lower P/S Ratio:** A lower P/S ratio suggests that investors are paying less for each SAR 1 of sales, implying lower growth potential or higher risk.

EV/EBITDA Ratio

Some companies do not pay dividends nor are they profitable. In this case, use the price-sales ratio, calculated as the price per share on the stock divided by sales per share, or the EV/EBITDA ratio.

EV: Enterprise Value represents the total market value of a company, including equity and debt. It provides a broader picture of a company's worth than market capitalization alone.

EBITDA: Earnings Before Interest, Taxes, Depreciation, and Amortization measures a company's operating performance, excluding certain non-operating expenses. It allows for comparison of profitability across industries and capital structures.

Maaden's EBITDA

500 million SAR

Industry Benchmark EV/EBITDA Ratio

8

Formula

$$(\text{EV} = \text{Benchmark EV/EBITDA Ratio} \times \text{EBITDA})$$

Calculation

$$(8 \times 500 = 4,000, million, SAR)$$

Sustainable Growth Rate

The Sustainable Growth Rate (SGR) is the maximum rate at which a company can grow its sales, earnings, and dividends without needing to raise additional capital. It represents the growth rate that can be achieved while maintaining the company's current financial structure and risk profile.

The SGR can be calculated using the following formula:

$$\text{Sustainable Growth Rate (SGR)} = \frac{ROE \times b}{1 - (ROE \times b)}$$

Where:

ROE measures a company's profitability in relation to shareholders' equity. It indicates how effectively a company uses its equity base to generate profits

The retention ratio, often denoted as (**b**), is the fraction of net income that is retained in the company instead of being paid out as dividends.

Sustainble Growth Key Components:

1 Retention Ratio (b)

The retention ratio is the percentage of earnings kept by a company instead of paying out dividends. A higher retention ratio means more money is reinvested for growth.

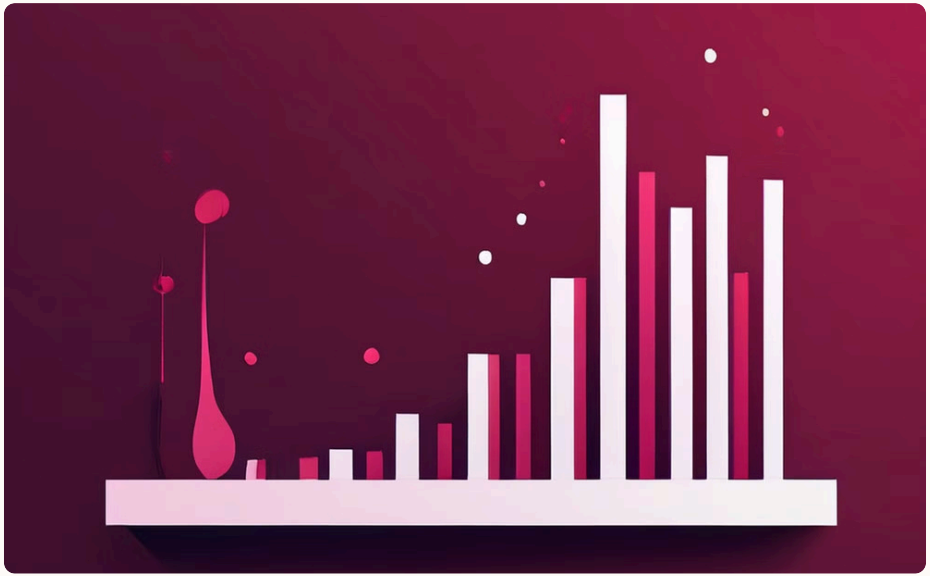
$$\text{Retention Ratio}(b) = 1 - \text{Dividend Payout Ratio} = 1 - \left(\frac{\text{Dividends Paid}}{\text{Net Income}}\right)$$

2 Return on Equity (ROE)

ROE measures how effectively a company uses shareholder's equity to generate profits. A higher ROE indicates more efficient use of equity.

$$ROE = Return\ on\ Equity = \frac{Net\ Income}{Shareholdr's\ Equity}$$

Key Takeaways



EV/EBITDA Ratio

The EV/EBITDA ratio is an important metric to consider when valuing a company. A higher EV/EBITDA ratio indicates that the market is willing to pay a premium for the company's future earnings potential.

Sustainable Growth Rate

The sustainable growth rate provides insight into a company's long-term growth potential. It represents the maximum rate at which a company can grow without needing to raise additional capital.

5.3
Stock Valuation

Principles of Finance

LO3: Lay out the different ways corporate directors are elected to office

Chapter 5

Cumulative Voting

This voting system is used in Saudi Arabia, where it gives minority shareholders greater power. With cumulative voting, you can combine all of your votes and cast them for a single director, increasing your chances of getting your preferred candidate elected.

Alinma Bank Example

For example, if you own 1,000 shares in Alinma Bank, and there are 3 directors up for election, you have 3,000 votes (1,000 shares x 3 directors). You can allocate these votes to one or more candidates.

Straight Voting

Each share gets one vote for each director position. This system makes it harder for minority shareholders to elect a director because they need to get a majority vote for each individual position.

Alinma Bank Example

100 shares = 100 votes per position. Favors majority shareholders.

Proxy Voting

Proxy voting allows shareholders to vote on company matters, like electing directors or approving proposals, without attending the annual shareholder meeting. Shareholders can delegate their voting rights to a proxy, often a company official or advocacy group.

Staggered Board

Some companies elect their directors in groups over a few years. This makes it harder for a group of investors to quickly take control of the board.

Petro Rabigh Example

Petro Rabigh has 9 directors, 3 elected each year. This creates stability and continuity on the board.

Strengths and Weaknesses

1. **Cumulative Voting**
   Allows minority influence, but can lead to strategic voting

2. **Straight Voting**
   Simple, but favors the majority shareholders

3. **Proxy Voting**
   Increases participation, but can lead to proxy battles

4. **Staggered Board**
   Provides stability, but can lead to management entrenchment

Wrap Up: Your Voice in Corporate Governance

Shareholder Rights

Shareholders have important rights that give them a say in how companies are run.

Voting Systems

Different voting systems impact shareholder power and influence over company decisions.

Proxy Voting

Proxy voting empowers shareholders to vote even when absent from meetings, giving them a voice.

Importance of Rights

These rights ensure corporate accountability, protect your investment, and influence management.

5.4
Stock Valuation

Principles of Finance

Define how the stock markets work – Chapter 5

Understanding the Tadawul

Connecting Investors and Companies

The Tadawul, Saudi Arabia's stock exchange, connects investors with companies seeking capital.

A Diverse Market

It's home to over 200 publicly listed companies across finance, energy, retail, technology, and more.

Two Key Markets

The Tadawul operates two key markets, catering to different stages of company growth.

Main Market and Nomu

The Main Market features established companies with a strong track record, while the Nomu, or Parallel Market, is tailored for smaller, innovative companies with a focus on growth.

Market Participants

Individual Investors

Institutional Investors

Market Makers

Brokers

Types of Orders

Market Order	**Immediate execution.**
Limit Order	**Specific price or better.**
Stop Order	**Triggered at a specific price.**

Market Capitalization Explained

Market Capitalization: This is the total value of a company's outstanding shares. It's calculated by multiplying the company's share price by the total number of shares. It gives you an idea of the company's overall size and value.

- **Example:** If a company has 10 million shares outstanding and each share is trading at SAR 50, its market capitalization is SAR 500 million (10 million x SAR 50).

Large Cap

Market cap over SAR 20 billion. Generally considered more stable, like **Aramco**, **SABIC** and **Al Rajhi Bank**.

Mid Cap

Market cap between SAR 10-20 billion. Offer growth potential with moderate risk.

Like **Jarir and Dar Al arkan.**

Small Cap

Market cap under SAR 2 billion like **Alhokair, Tabuk Cement**. Higher growth potential but riskier.

Dividend Yield: Income from Stocks

Definition

Dividend Yield represents the annual dividend payment as a percentage of the current stock price. It shows how much you can expect to earn in dividends relative to the stock's value.

Example

If a company pays a SAR 2 annual dividend and its share price is SAR 50, the dividend yield is 4% (SAR 2 / SAR 50 x 100).

Importance

Dividend yield is crucial for income-focused investors.

Earnings Per Share (EPS): Profitability Indicator

EPS measures a company's profitability on a per-share basis. It's calculated by dividing net income by outstanding shares.

A higher EPS often indicates stronger financial performance.

Net Income	Outstanding Shares	EPS
SAR 100 million	10 million	SAR 10
SAR 50 million	5 million	SAR 10
SAR 200 million	40 million	SAR 5

Price to Earnings Ratio: Valuation Metric

15

P/E Ratio

Indicates how much investors are willing to pay for earnings.

10

Higher P/E

May suggest high growth expectations or overvaluation.

5

Lower P/E

May indicate undervaluation or slower growth.

Compare P/E ratios within the same industry for context.

If a stock trades at SAR 60 and has an EPS of SAR 4, its P/E ratio is 15 (60 / 4 = 15). This means investors are willing to pay 15 times the company's earnings for the stock.

Key Takeaways

Market Capitalization

Indicates the overall value of a company based on its outstanding shares.

Dividend Yield

Shows how much you can expect to earn in dividends relative to the stock's price.

Earnings Per Share (EPS)

Measures a company's profitability on a per-share basis.

Price to Earnings Ratio (P/E Ratio)

Compares a company's stock price to its earnings per share, offering insights into valuation.